Adaptive Trace-On Replay Experiment
Mozilla Firefox-Android Telemetry

End-to-End Reproducible Experiment -- Journal Paper Supplement

March 2026

1. Experiment Overview

This report documents a reproducible end-to-end replay experiment that evaluates four classical adaptive anomaly detectors as gatekeepers for a performance-telemetry collection system. In each simulated scenario the detector's binary decision output (decision = 1 => trace ON) determines which data points are recorded. Three metrics are measured consistently across all predictors on the chronological test portion of each time series:

(1) Storage Reduction -- fraction of test-set points NOT recorded.
(2) Trigger Frequency -- number of trace-on activations per test time step (sensor switching overhead).
(3) Alert Coverage -- fraction of ground-truth regression alerts covered by a trace-on interval (+/-tau pushes).

Dataset and Experimental Settings

Dataset:	Mozilla Firefox-Android Treeherder telemetry
Predictors:	ARIMA, LAST, SMA, EWMA
Total signatures:	214
Alert tolerance tau:	+/-5 pushes
Train / Test split:	70% / 30%

Algorithm 3 Trace-On Logic

At each test-set time step t:

1. Compute forecast y_hat_t using the chosen predictor.
2. Compute residual delta_t = |y_t - y_hat_t|.
3. Compute dynamic threshold Theta_t = max(Theta_min, k_sigma * std(delta_{t-w:t})).
4. Update anomaly score: s_t = beta * 1[delta_t > Theta_t] + (1-beta)*s_{t-1}.
5. Emit decision_t = 1 (trace ON) if s_t >= theta_anomaly, else 0 (trace OFF).

Storage overhead is determined by the density of decision=1 steps. Trigger frequency counts how often the system switches from OFF to ON, which represents the cost of activating the sensor. Alert coverage measures whether real regression events were captured.

2. Aggregate Results

Method	Storage Red.	Trigger Freq.	Alert Cov.	Precision
ARIMA(1,1,1)	89.8%	58.229e-3	85.7%	0.086
LAST (naive)	89.4%	56.519e-3	70.2%	0.064
SMA(w=20)	88.8%	57.612e-3	93.0%	0.092

EWMA(a=0.30)	91.0%	60.710e-3	79.0%	0.073

Storage Reduction: fraction of test-set points suppressed (higher is better for storage savings). Trigger Frequency (x10^-3): activations per step (lower means fewer sensor switches, less overhead). Alert Coverage: recall of regression alerts on the test set (higher means more regressions caught). Precision: fraction of detected intervals that coincide with a real alert.

Storage Reduction -- Detailed Statistics

Method	Mean	Std	Median	n_activate	Burst Ratio
ARIMA(1,1,1)	89.8%	7.3%	90.8%	10.3	0.515
LAST (naive)	89.4%	3.0%	89.6%	10.0	0.632
SMA(w=20)	88.8%	6.0%	89.9%	10.2	0.574
EWMA(a=0.30)	91.0%	2.7%	91.1%	10.7	0.453

Burst Ratio: fraction of trace-on time that occurs in contiguous runs longer than one step. A high burst ratio means the detector tends to stay ON for extended periods rather than flickering, which is operationally preferable as it reduces the number of start/stop events.

3. Figures

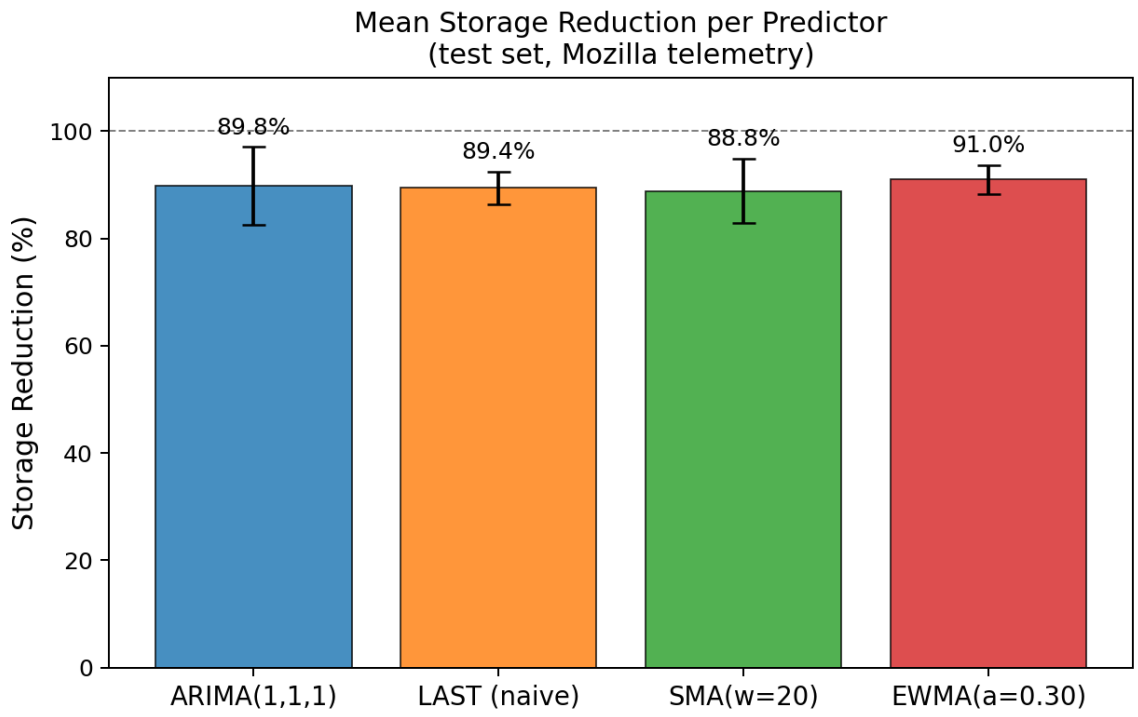


Figure 1 - Mean storage reduction per predictor (mean +/- std, test set).

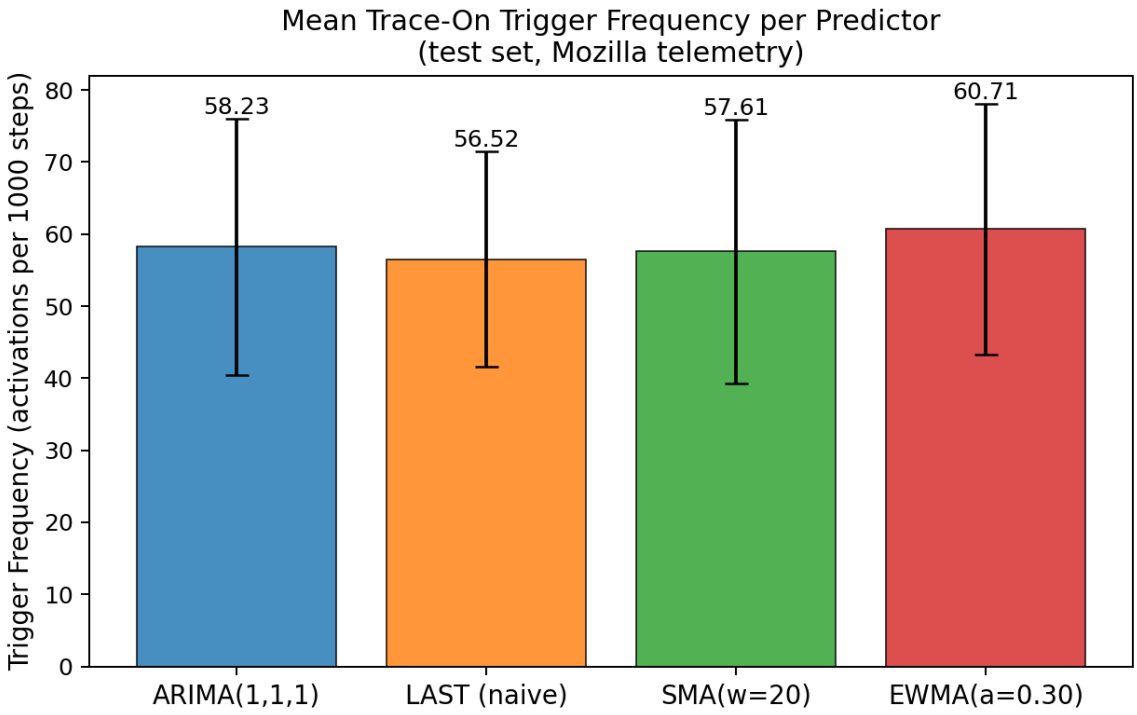


Figure 2 - Mean trigger frequency per predictor (activations per 1000 test steps).

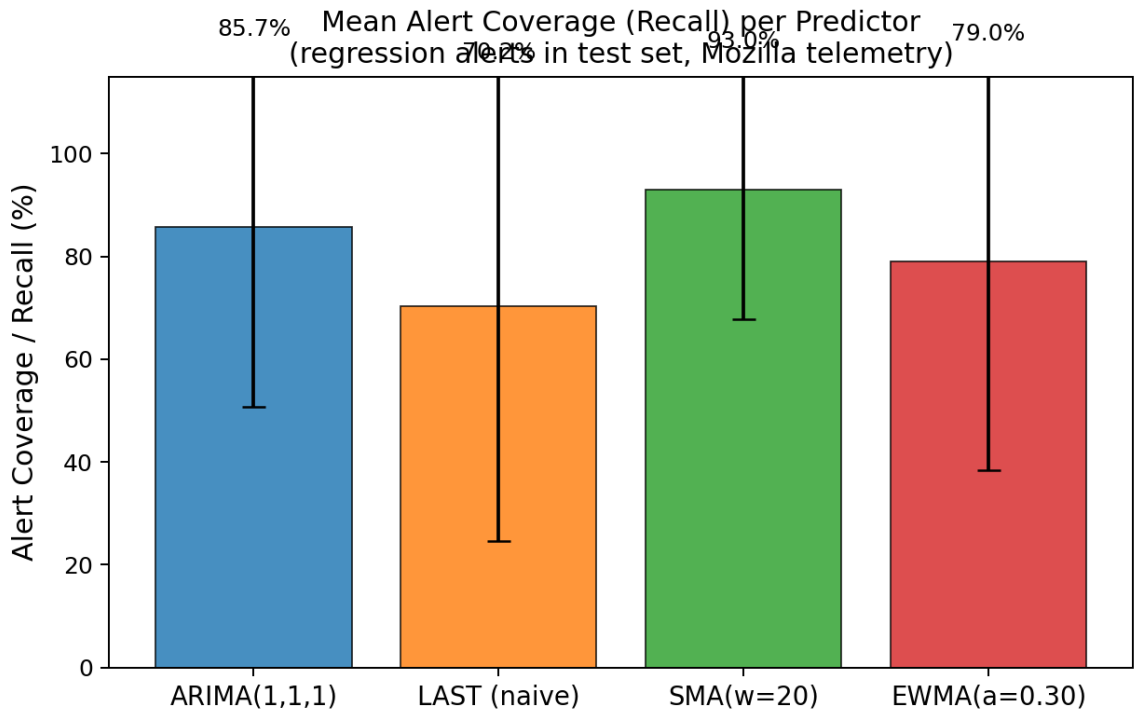
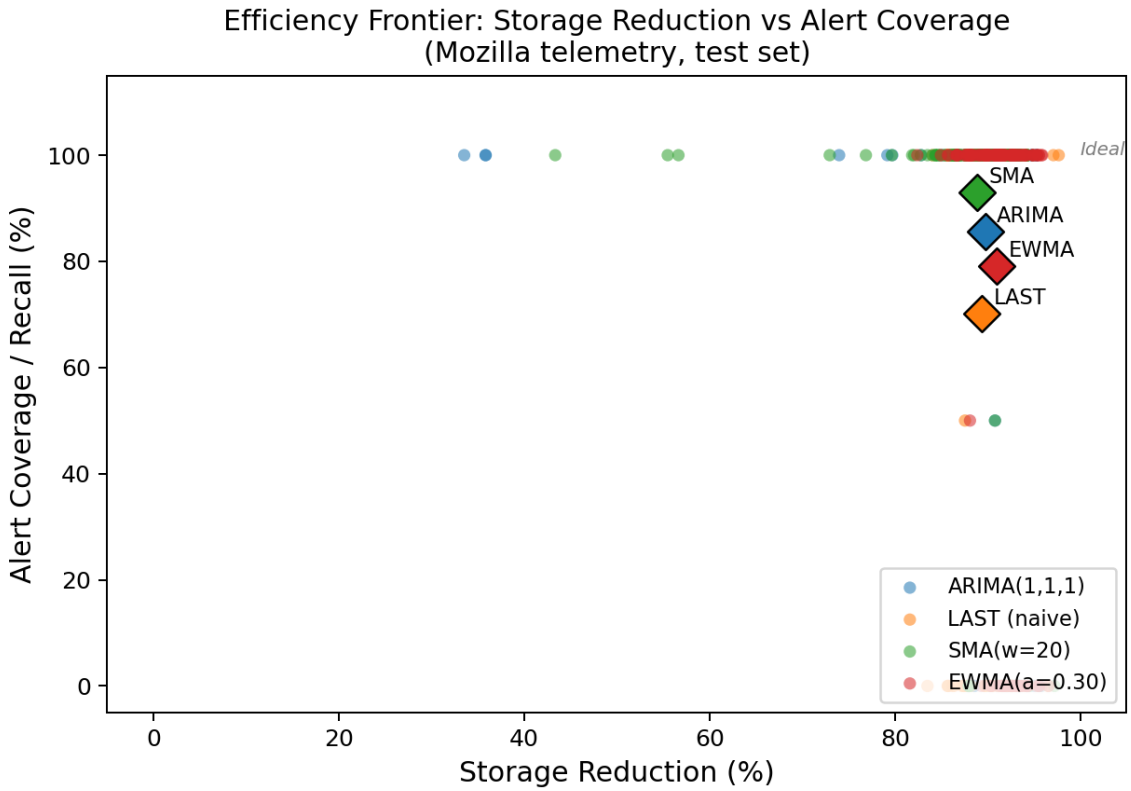


Figure 3 - Mean alert coverage (recall) per predictor on the test set.



4 - Efficiency frontier: each dot is one signature. Diamond markers show per-method centroids. Top-right corner is ideal (high coverage, high storage s

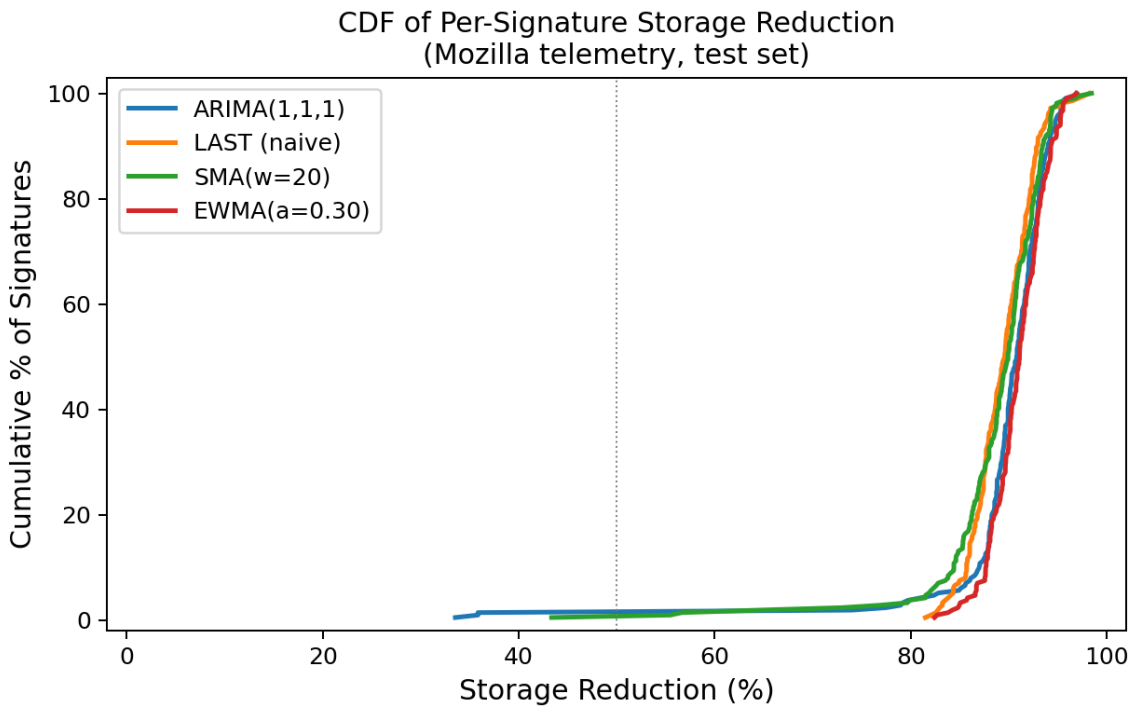


Figure 5 - CDF of per-signature storage reduction. A curve shifted to the right indicates higher storage savings.

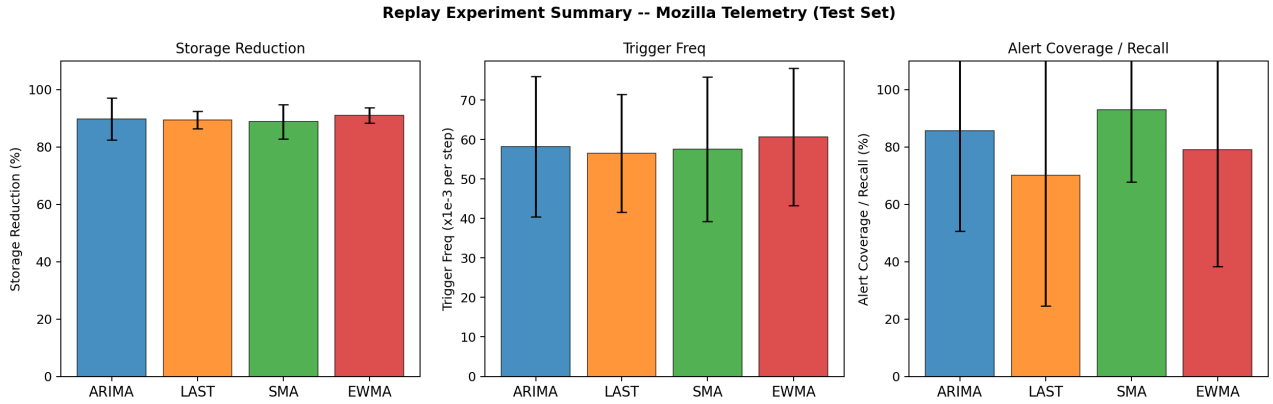


Figure 6 - Summary panel: all three metrics side by side.

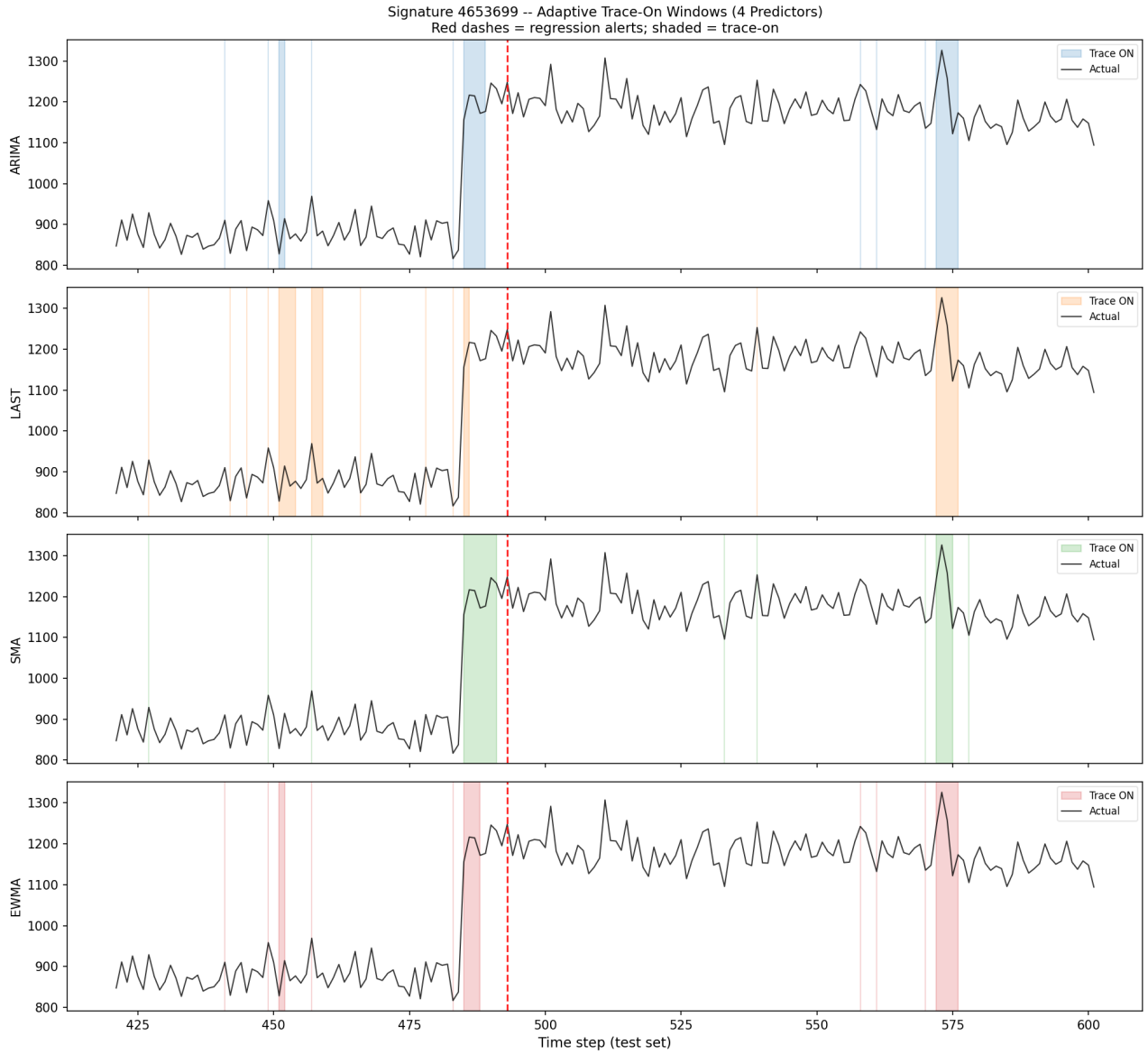


Figure 7 - Sample timeline for signature 4653699. Shaded regions = trace ON. Red dashes = regression alert events.

4. Per-Signature Results (Top 15 by Storage Reduction)

Method: ARIMA(1,1,1)

Signature	T_test	Stor.Red.	Trg.Freq.	Activations	Cov.	Precision

4969038	65	96.9%	30.769	2	---	0.000
4654361	157	96.8%	31.847	5	---	0.000
4653882	191	95.8%	31.414	6	0.0%	0.000
4654024	180	95.6%	38.889	7	0.0%	0.000
4669835	782	95.4%	42.199	33	---	0.000
4653828	171	95.3%	35.088	6	100.0%	0.167
4653829	171	95.3%	23.392	4	100.0%	0.250
4653838	171	95.3%	46.784	8	---	0.000
4653771	169	95.3%	47.337	8	100.0%	0.125
4654027	180	95.0%	44.444	8	0.0%	0.000
4654584	176	94.9%	39.773	7	100.0%	0.286
4654387	173	94.8%	40.462	7	100.0%	0.286
4653881	191	94.8%	52.356	10	---	0.000
4653887	191	94.8%	47.120	9	---	0.000
4653825	171	94.7%	40.936	7	100.0%	0.143

Method: LAST (naive)

Signature	T_test	Stor.Red.	Trg.Freq.	Activations	Cov.	Precision
4653782	169	98.2%	17.751	3	---	0.000
4654564	173	97.7%	23.121	4	100.0%	0.250
4654568	173	97.1%	28.902	5	100.0%	0.200
4654567	173	96.5%	34.682	6	0.0%	0.000
4669835	782	95.4%	42.199	33	---	0.000
4969038	65	95.4%	30.769	2	---	0.000
4654351	157	94.3%	44.586	7	100.0%	0.286
4654352	157	94.3%	44.586	7	100.0%	0.286
4654042	169	94.1%	59.172	10	100.0%	0.200
4670242	184	94.0%	43.478	8	---	0.000
4670243	184	94.0%	43.478	8	---	0.000
4653700	181	93.9%	44.199	8	0.0%	0.000
4969001	64	93.8%	62.500	4	100.0%	0.250
4654566	173	93.6%	57.803	10	100.0%	0.100
4654350	157	93.6%	50.955	8	100.0%	0.250

Method: SMA(w=20)

Signature	T_test	Stor.Red.	Trg.Freq.	Activations	Cov.	Precision
4969038	65	98.5%	15.385	1	---	0.000
4654026	180	97.2%	27.778	5	0.0%	0.000
4654361	157	96.8%	31.847	5	---	0.000
4653708	181	95.6%	44.199	8	---	0.000
4654585	176	94.9%	28.409	5	100.0%	0.200
4654678	175	94.9%	40.000	7	---	0.000
4654024	180	94.4%	33.333	6	0.0%	0.000
4654025	180	94.4%	38.889	7	0.0%	0.000
4969022	71	94.4%	56.338	4	---	0.000
4654583	176	94.3%	34.091	6	100.0%	0.333

4653987	175	94.3%	28.571	5	100.0%	0.200
4653988	175	94.3%	28.571	5	100.0%	0.200
4653998	175	94.3%	57.143	10	---	0.000
4653789	174	94.2%	28.736	5	100.0%	0.200
4653881	191	94.2%	52.356	10	---	0.000

Method: EWMA(a=0.30)

Signature	T_test	Stor.Red.	Trg.Freq.	Activations	Cov.	Precision
4969038	65	96.9%	30.769	2	---	0.000
4654583	176	96.6%	22.727	4	0.0%	0.000
4653825	171	95.9%	40.936	7	100.0%	0.143
4653883	191	95.8%	41.885	8	100.0%	0.125
4654024	180	95.6%	38.889	7	0.0%	0.000
4654026	180	95.6%	44.444	8	0.0%	0.000
4654351	157	95.5%	31.847	5	100.0%	0.400
4654352	157	95.5%	31.847	5	100.0%	0.400
4653828	171	95.3%	35.088	6	100.0%	0.167
4653829	171	95.3%	35.088	6	100.0%	0.167
4653881	191	95.3%	47.120	9	---	0.000
4653882	191	95.3%	36.649	7	0.0%	0.000
4653887	191	95.3%	47.120	9	---	0.000
4653771	169	95.3%	47.337	8	100.0%	0.125
4654348	157	94.9%	38.217	6	100.0%	0.333

5. Discussion and Reproducibility

Reproducibility: This experiment is fully deterministic given the pre-computed detector output CSVs (mozilla_results*.csv) produced by run_pipeline.py. All random seeds are fixed by the deterministic nature of the Algorithm 3 controller. Re-running 11_replay_experiment.py from the workspace root will reproduce all tables, plots, and this PDF identically.

Storage vs Coverage Trade-off: A high storage reduction with high alert coverage is the ideal operating point. Detectors that achieve both demonstrate that the adaptive gating strategy correctly identifies anomalous windows without recording excessive normal operation data.

Trigger Frequency: Low trigger frequency is desirable in practice because each ON activation involves system overhead (e.g., starting a logger, allocating buffers). Methods that activate infrequently but cover all alerts are particularly suitable for resource-constrained deployments.

Limitations: Alert coverage is limited to the test-set portion of each time series (first 70% is training). Signatures without ground-truth regression alerts in the test set contribute to storage/trigger metrics but are excluded from alert-coverage aggregation (reported separately as 'n_alerted_with_cov' in replay_summary.json).

6. Output File Index

File	Description
results/replay/replay_results.csv	Per-signature x method replay metrics

results/replay/replay_summary.json	Aggregate statistics (JSON)
results/replay/plots/replay_storage_reduction.png	Figure 1 -- bar chart
results/replay/plots/replay_trigger_frequency.png	Figure 2 -- bar chart
results/replay/plots/replay_alert_coverage.png	Figure 3 -- bar chart
results/replay/plots/replay_efficiency_frontier.png	Figure 4 -- scatter
results/replay/plots/replay_cdf_storage.png	Figure 5 -- CDF
results/replay/plots/replay_summary_panel.png	Figure 6 -- summary panel
results/replay/plots/replay_timeline_sample.png	Figure 7 -- timeline
results/replay/replay_report.pdf	This PDF report

Generated automatically by Code/11_replay_experiment.py -- March 2026